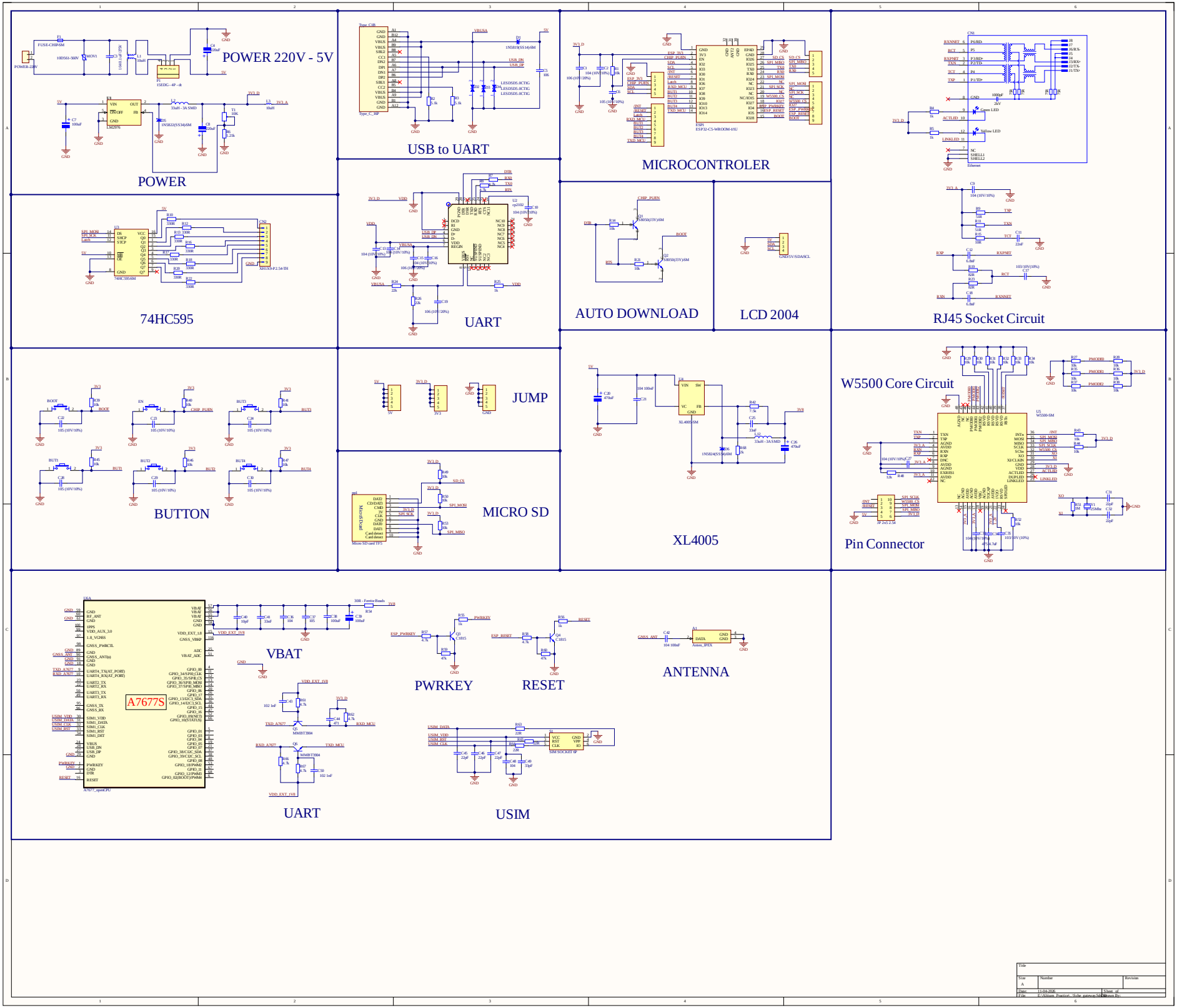

POWER 220V - 5V
POWER
USB to UART
MICROCONTROLER
74HC595
UART
AUTO DOWNLOAD
LCD 2004
RJ45 Socket Circuit
BUTTON
JUMP
MICRO SD
XL4005
W5500 Core Circuit
Pin Connector
VBAT
PWRKEY
RESET
ANTENNA
UART
USIM

| Comment | Description | Designator | Footprint | LibRef | Quantity |
|---|---|---|---|---|---|
| Anten_IPEX | | A1 | Anten_IPEX | Anten_IPEX | 1 |
| BOOT | Nut Nhan 4 Chan 6x6mm | BT1 | Nut Nhan 4 Chan Cam 6x6 | Nut Nhan 4 Chan Cam 6x6 | 1 |
| EN | Nut Nhan 4 Chan 6x6mm | BT2 | Nut Nhan 4 Chan Cam 6x6 | Nut Nhan 4 Chan Cam 6x6 | 1 |
| BUT3 | Nut Nhan 4 Chan 6x6mm | BT3 | Nut Nhan 4 Chan Cam 6x6 | Nut Nhan 4 Chan Cam 6x6 | 1 |
| BUT1 | Nut Nhan 4 Chan 6x6mm | BT4 | Nut Nhan 4 Chan Cam 6x6 | Nut Nhan 4 Chan Cam 6x6 | 1 |
| BUT2 | Nut Nhan 4 Chan 6x6mm | BT5 | Nut Nhan 4 Chan Cam 6x6 | Nut Nhan 4 Chan Cam 6x6 | 1 |
| BUT4 | Nut Nhan 4 Chan 6x6mm | BT6 | Nut Nhan 4 Chan Cam 6x6 | Nut Nhan 4 Chan Cam 6x6 | 1 |
| 106 (10V/20%) | Dán | C1, C15, C19 | tu 0805 | Tu 0805 106 16V 10% | 3 |
| 104 (10V/10%) | Dán | C2, C9, C10, C13, C14, C16, C27, C33 | tu 0805 | Tu 0805 104 50V 10% | 8 |
| 104 0.1uF/275V | Dán | C3 | tu 0805 | Tu 0805 104 50V 10% | 1 |
| 220uF | Dán | C4, C8 | tu nhom 8 (e) | Tu Nhom 220uF 10V | 2 |
| 106 | Dán | C5 | tu 0805 | Tu 0805 106 16V 10% | 1 |
| 105 (10V/10%) | Dán | C6, C22, C23, C24, C28, C29, C30 | tu 0805 | Tu 0805 105 50V 10% | 7 |
| 100uF | Dán | C7 | tu nhom 8 (f) | Tu Nhom 100uF 50V | 1 |
| 22nF | Dán | C11 | tu 1206 | Tu 1206 22pF 50V 10% | 1 |
| 6.8nF | Dán | C12, C18 | tu 0805 | Tu 0805 680pF 50V 10% | 2 |
| 103/10V(10%) | Dán | C17 | tu 1206 | Tu 1206 104 50V 20% | 1 |
| 470uF | Dán | C20, C26 | tu nhom 10 (G) | Tu Nhom 470uF 35V | 2 |
| 104 100nF | Dán | C21, C42 | tu 0805 | Tu 0805 104 50V 10% | 2 |
| 33nF | Dán | C25, C41 | tu 0805 | Tu 0805 33nF 50V 20% | 2 |
| 22pF | Dán | C31, C32, C45, C46, C47 | tu 0805 | Tu 0805 22pF 50V 10% | 5 |
| 475 4.7uF | Dán | C34 | tu 0805 | Tu 0805 475 25V 20% | 1 |
| 103/10V (10%) | Dán | C35 | tu 0805 | Tu 0805 104 50V 10% | 1 |
| 104 | Dán | C36, C48 | tu 0805 | Tu 0805 104 50V 10% | 2 |
| 105 | Dán | C37 | tu 0805 | Tu 0805 105 50V 10% | 1 |
| 100uF | Dán | C38 | tu 0805 | Tu 0805 33nF 50V 20% | 1 |
| 100uF | Dán | C39 | tu nhom 6.3 (d) | Tu Nhom 100uF 25V | 1 |
| 10pF | Dán | C40 | tu 0805 | Tu 0805 10pF 50V 10% | 1 |
| 102 1nF | Dán | C43, C50 | tu 0805 | Tu 0805 102 50V 10% | 2 |
| 471 | Dán | C44 | tu 0805 | Tu 0805 471 50V 10% | 1 |
| 33pF | Dán | C49 | tu 0805 | Tu 0805 33pF 50V 10% | 1 |
| Ethernet | Single Port RJ45 Connector With Magnetics Module And LED - Through Hole | CN1 | RJ45-HR911105A-TH | RJ45-HR911105A-TH | 1 |
| XH1X9-P2.54-TH | Connector XH2.54 9P - Through Hole | CN2 | XH1X9M-P2.54-TH | XH1X9-P2.54-TH | 1 |
| 1N5819(SS14)-SM | 1.0A Schottky Barrier Rectifiers - Surface Mount | D1 | SMA-J_DO-214AC | 1N5819(SS14)-SM | 1 |
| LESD5D5.0CT1G | | D2, D3, D4 | LESD5D5.0CT1G, | LESD5D5.0CT1G | 3 |
| 1N5822(SS34)-SM | 3.0A Schottky Barrier Rectifiers - Surface Mount | D5 | SMA-J_DO-214AC | 1N5822(SS34)-SM | 1 |
| 1N5824(SS54)-SM | 5.0A Schottky Barrier Rectifiers - Surface Mount | D6 | SMA-J_DO-214AC | 1N5824(SS54)-SM | 1 |
| ESP32-C5-WROOM-1/1U | | ESP1 | ESP32-C5-WROOM-1/1U | ESP32-C5-WROOM-1/1U | 1 |
| FUSE-CHIP-SM | Chip Fuses - Surface Mount | F1 | FUSE-CHIP-1812-SM | FUSE-CHIP-SM | 1 |
| POWER-220V | | HB1 | HB9500 | HB9500 | 1 |
| JP esp | Cắm | JP1, JP2 | JP 1x5 DT | JP 1x5 2.54 Duc Thang | 2 |
| JP esp | Cắm | JP3, JP4 | JP 1x9 DT | JP 1x9 2.54 Duc Thang | 2 |
| GND/5V/SDA/SCL | Cắm | JP5 | JP 1x4 DT | JP 1x4 2.54 Duc Thang | 1 |
| 5V | Cắm | JP6 | JP 1x5 DT | JP 1x5 2.54 Duc Thang | 1 |
| 3V3 | Cắm | JP7 | JP 1x5 DT | JP 1x5 2.54 Duc Thang | 1 |
| GND | Cắm | JP8 | JP 1x5 DT | JP 1x5 2.54 Duc Thang | 1 |
| JP 2x5 2.54 | Cắm | JP9 | JP 2x5 DT | JP 2x5 2.54 Duc Thang | 1 |
| 10uH | EMI Filter Inductor - Through Hole | L1 | L-UU9.8-L17.0-W12.0-H17.0-TH | L-FILTER-TH | 1 |
| 33uH - 3A SMD | Dán | L2, L12 | Cuon Cam 12x12x10 | CUON CAM 100uH SMD | 2 |
| 10uH | Chip Inductor - Surface Mount | L3 | L-CHIP-0805-T1.05-SM | L-CHIP-SM | 1 |
| 10D561-560V | Metal Oxide Varistor - Through Hole | MOV1 | MOV-P7.5-D17.0-T3.6-TH | MOV-TH | 1 |
| 15EDG - 4P - dt | 15EDG - 4P - 3.81 mm đực thẳng | P1 | 15EDG - 4P - dt | 15EDG - 4P - dt | 1 |
| Micro SD card TF5 | | pp1 | MicroSD TF V5 | Micro SD card TF5 | 1 |
| S8050(J3Y)-SM | 0.5A 25V NPN General Purpose Transistors - Surface Mount | Q1, Q2 | SOT23-3_TO-236AB | S8050(J3Y)-SM | 2 |
| C1815 | NPN transistor | Q3, Q4 | SOT23 | C1815 HF SMD | 2 |
| MMBT3904 | NPN General Purpose Transistors 0.2A 40V - Surface Mount | Q5, Q6 | SOT23-3_TO-236AB | 2N3904(1AM)-SM | 2 |
| 10k | Dán | R1, R14, R21, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R43, R44, R45, R46, R47, R49, R50, R52, R53 | tro smd 0805 | Tro 0805 5% 10k | 27 |
| 5.1k | Dán | R2, R3 | tro smd 0805 | Tro 0805 1% 5.1k | 2 |
| 1k | Dán | R4, R5, R25, R55, R56 | tro smd 0805 | Tro 0805 5% 1k, Tro 0805 1% 1k | 5 |
| 1.21k | Dán | R6 | tro smd 1206 | Tro 1206 5% 1k | 1 |
| 4.7k | Dán | R7, R8, R57, R58, R61, R62, R66, R67 | tro smd 0805 | Tro 0805 5% 4.7k | 8 |
| 51R | Dán | R9, R11 | tro smd 0805 | Tro 0805 5% 30R | 2 |
| 330R | Dán | R10, R12, R13, R16, R17, R18, R20, R22 | Tro SMD 0805 | Tro 0805 5% 330R | 8 |
| 10R | Dán | R15 | tro smd 0805 | Tro 0805 5% 10R | 1 |
| 62R | Dán | R19, R23 | tro smd 0805 | Tro 0805 5% 62R | 2 |
| 22k | Dán | R24, R26 | tro smd 0805 | Tro 0805 5% 22k | 2 |
| 7.5k | Dán | R42 | tro smd 0805 | Tro 0805 5% 4.7k | 1 |
| 12k | Dán | R48 | tro smd 0805 | Tro 0805 5% 12k | 1 |
| 1M | Dán | R51 | tro smd 0805 | Tro 0805 1% 1M | 1 |
| 39R - Ferrite Beads | Dán | R54 | Tro SMD 0805 | Tro 0805 5% 390R | 1 |
| 47k | Dán | R59, R60 | tro smd 0805 | Tro 0805 5% 47k | 2 |
| 22R | Dán | R63, R64, R65 | tro smd 0805 | Tro 0805 5% 22R | 3 |
| 2k | Dán | R68 | tro smd 0805 | Tro 0805 1% 2k | 1 |
| SIM SOCKET 6P | | S1 | SIM SOCKET 6P | SIM SOCKET 6P | 1 |
| 10K | Cắm | T1 | Bien Tro Vi Chinh Dung | Bien Tro Tinh Chinh 10K Dung | 1 |
| Type_C_16P | | Type_C1 | Type_C_16P | Type_C_16P | 1 |
| LM2976 | | U1 | TO263 | LM2596 | 1 |
| cp2102 | | U2 | cp2102 | cp2102 | 1 |
| 74HC595-SM | 8-Bit Shift Registers With 3-State Output Registers - Surface Mount | U3 | TSSOP16-0.65-4.4MM | 74HC595-SM | 1 |
| XL4005-SM | 5A 180KHz 36V Buck DC to DC Converter - Surface Mount | U4 | D2PAK-6_TO-263BA | XL4015-SM | 1 |
| W5500-SM | Ethernet ICs 3in1 Enet Controller TCP/IP +MAC+PHY - Surface Mount | U5 | LQFP48-0.5-7X7MM | W5500-SM | 1 |
| A7677_openCPU | LTE CAT 1 module, 4G + (GNSS) | U6 | A7677 | A7677_openCPU | 1 |
| 25Mhz | Crystal For all Crystal 3pin. | Y1 | Thach Anh 2 chan cam | Thach Anh 2 chan Cam | 1 |